

[Angles directed by default]

Problem 1 (22AIME11)

Let the circle touch AB, BC, AD at X, Y, Z respectively. Note that $BC \perp YZ \perp AD$ due to tangency. Additionally, $AX = 6$ and $CY = 20$ using Power of a Point. By Pythagorean's on $ACZY$, we know the length $YZ = 6\sqrt{3}$. If we let the length of $BX = BY = m$, we may get, dropping an altitude from B to AD then using Pythagorean's,

$$(6 - m)^2 + 108 = (6 + m)^2 \Leftrightarrow 24m = 108 \Leftrightarrow m = \frac{9}{2}$$

The answer is,

$$\left(20 + \frac{9}{2}\right) \cdot 6\sqrt{3} = 147\sqrt{3}$$

Problem 3 (23AIME8)

Let the projections of P onto AB, BC, CD, DA be W, X, Y, Z respectively. Note that, since $ABCD$ is a rhombus, using area argument,

$$PX + PZ = PW + PY \Leftrightarrow PY = 20$$

Let $PY \cap AD = T$. By POP,

$$ZT^2 = 9 \cdot 16 = 144 \Leftrightarrow ZT = 12$$

Therefore, $PT = 15$ by the Pythagorean Theorem. By similar triangles,

$$\frac{AD}{WY} = \frac{PT}{ZT} = \frac{5}{4}$$

Therefore, $AD = 25 \cdot \frac{5}{4} = \frac{125}{4}$. The perimeter is hence 125.

4 clubs; 1.5 hours

Problem 6 (16ARMLOC)

Lemma: In a triangle ABC with length of A -median d , $b^2 + c^2 = 2d^2 + \frac{a^2}{2}$.

Proof: We use Stewart's to get,

$$\frac{a^3}{4} + ad^2 = \frac{a}{2} \cdot b^2 + \frac{a}{2} \cdot c^2 \Leftrightarrow b^2 + c^2 = 2d^2 + \frac{a^2}{2} \quad \square$$

We may plug this in on ABP, BPC, CPD to get,

$$AP^2 + BP^2 = 2 \cdot 5^2 + \frac{4^2}{2} = 58$$

$$BP^2 + CP^2 = 2 \cdot 7^2 + \frac{6^2}{2} = 116$$

$$CP^2 + DP^2 = 2 \cdot 9^2 + \frac{8^2}{2} = 194$$

As a result,

$$AP^2 + DP^2 = 136$$

If we let O be the center of the circle, note that by symmetry, BE, CF, AD are diameters. Therefore, applying the lemma on APD, BPE, CPF , we see that,

$$AP^2 + DP^2 = BP^2 + EP^2 = CP^2 + FP^2 = 136$$

Therefore,

$$EP^2 + FP^2 = (BP^2 + EP^2) + (CP^2 + FP^2) - (BP^2 + CP^2) = 156$$

Equivalently,

$$2PK^2 + \frac{FE^2}{2} = 156$$

$$\Leftrightarrow PK^2 = 69$$

Problem 8 (10AIME15)

Let the incircles of ABM, ACM , and ABC touch BC at X_B, X_C, X respectively. Consider the homotheties at B and C sending the incircle of ABC to that of ABM and ACM . Since the incircles have same radii, the same scale k is used. Since $BX = 7$ and $CX = 8$, we have $BX_B = 7k$ and $CX_C = 8k$. Let d be the length of AM . Note that $8t + d$ is the semi-perimeter of AMC and $7t + d$ is that of AMB . Therefore, the combined perimeter of both triangles is,

$$2(d + 7k) + 2(d + 8k) = 4d + 30k$$

However, this would also be equal to $12 + 13 + 15 + 2d$.

$$4d + 30k = 12 + 13 + 15 = 2d$$

$$\Leftrightarrow d = 20 - 15k$$

We may get that the semi-perimeter of ABM is $20 - 15k + 7k = 20 - 8k$. Therefore, $MX_B = s - 12 = 8 - 8k \Leftrightarrow BM = 8 - k \Leftrightarrow CM = 7 + k$. We apply Stewart's theorem to solve for k and finish,

$$13^2(8 - k) + 12^2(7 + k) = (8 - k)(7 + k)(15) + (20 - 15k)^2(15)$$

$$\Leftrightarrow k = \frac{2}{3}, 2$$

The solution that $k = 2$ is degenerate as that would result in $d < 0$. Therefore, $k = \frac{2}{3}$ and we have

$BM = \frac{22}{3}; CM = \frac{23}{3}$. The answer is $\frac{22}{23}$.

14 clubs; 4 hours

Problem 10 (22BISSUE13)

Translate the problem in the $-x$ direction by 23 units. Since line l passes through $(0, u)$, it passes through the x -axis at $C = (-13u, 0)$ and has equation $y = \frac{1}{13}x + u$.

Let the cubic have equation $\frac{u}{3mn}(x+3)(x-m)(x-n)$ such that it passes through $A, B, (-3, 0)$, and $(0, u)$ with $A = (m, 0)$ and $B = (n, 0)$. We know, where directed lengths are in use,

$$CA \cdot CB = (-13u - m)(-13u - n)$$

Let p, q be the projections of P, Q onto the x -axis. Since, by the Pythagorean theorem, the distance between two points on l whose projection onto the x -axis are d apart is $d \cdot \frac{\sqrt{170}}{13}$. Where directed lengths are in use,

$$CQ \cdot CP = \left(\frac{\sqrt{170}}{13}\right)^2 (-13u - p)(-13u - q) = \frac{170}{169}(-13u - p)(-13u - q)$$

The cubic also intersects l at,

$$\frac{u}{3mn}(x+3)(x-m)(x-n) - \frac{1}{13}x - u = 0$$

Therefore,

$$\frac{u}{3mn}(x)(x-p)(x-q) = \frac{u}{3mn}(x+3)(x-m)(x-n) - \frac{1}{13}x - u$$

At $x = -13u$, we have,

$$(-13u)(-13u - p)(-13u - q) = (-13u + 3)(-13u - m)(-13u - n)$$

As a result,

$$CA \cdot CB = CQ \cdot CP$$

$$\Leftrightarrow (-13u - m)(-13u - n) = \frac{170}{169}(-13u - p)(-13u - q)$$

$$\Leftrightarrow (-13u - m)(-13u - n) = \frac{170}{169}(-13u - m)(-13u - n) \cdot \frac{(-13u + 3)}{(-13u)}$$

Therefore,

$$\frac{(-13u + 3)}{(-13u)} = \frac{169}{170} \Leftrightarrow u = \frac{510}{13}$$

Problem 11 (12HMMTG8)

We let $DE = x$; $AF = y$. Note that, by Pitot, $x + y + 6 = 9 + 4 + 2$ so $x + y = 9$.

By Brianchon, AD, BE, CF intersect at P .

We let $PF = kyx$ for some k . Note that, by similar triangles APF, CPD , we have $PD = 2kx$.

Similar triangles DPE, BPA gives $BP = 18k$.

Similar triangles BPC, FPE gives $PF = 12k$.

Therefore, $12k = kyx$ means $12 = xy$. We may get,

$$\{x, y\} = \left\{ \frac{9 \pm \sqrt{81 - 48}}{2} \right\} = \left\{ \frac{9 - \sqrt{33}}{2}, \frac{9 + \sqrt{33}}{2} \right\}$$

24 clubs; 5.5 hours

Problem 12 (09PUMAC)

We angle chase to get $\angle BMC = 100$. We let $\angle MCB = x$; $\angle MBC = 80 - x$. Using trigonometric Ceva's,

$$\begin{aligned} \frac{\sin(30)}{\sin(x)} \cdot \frac{\sin(80 - x)}{\sin(20)} \cdot \frac{\sin(10)}{\sin(40)} &= 1 \\ \Leftrightarrow \frac{\sin(x)}{\sin(80 - x)} &= \frac{\sin(10) \sin(30)}{\sin(20) \sin(40)} \\ \Leftrightarrow \frac{\sin(x)}{\sin(80 - x)} &= \frac{\sin(30)}{2 \cos(10) \sin(40)} \\ \Leftrightarrow \frac{\sin(x)}{\sin(80 - x)} &= \frac{\sin(30)}{\sin(50) + \sin(30)} \\ \Leftrightarrow \frac{\sin(x)}{\sin(80 - x)} &= \frac{1}{2 \sin(50) + 1} \\ \Leftrightarrow \sin(80 - x) - \sin(x) &= 2 \sin(x) \sin(50) \\ \Leftrightarrow 2 \cos(40) \sin(40 - x) &= 2 \sin(x) \sin(50) \end{aligned}$$

Since $\sin(50) = \cos(40)$,

$$\Leftrightarrow \sin(x) = \sin(40 - x)$$

Clearly, $x = 20$ on $0 < x < 80$ is the only solution. Therefore, $\angle MBC = 60$.

Problem 14 (20AIMEII15)

If we let M be the midpoint of BC , we have,

$$\angle XMT = \angle XBT = \angle ACB = 180 - \angle BCY = \angle YTM$$

Analogously, $\angle YMT = \angle XTM$. Therefore, $XYMT$ is a parallelogram and,

$$\begin{aligned} 2XM^2 + 2MY^2 &= MT^2 + XY^2 \\ \Leftrightarrow 2286 - 2MY^2 &= MY^2 + 135 \\ \Leftrightarrow 717 &= MY^2 \end{aligned}$$

The answer is hence **half of 1434**.

31 clubs; 8 hours

Problem 16 (2112A24)

Let O be the center of the semicircle and I be that of (PQR) .

Note that, by law of sines, $\frac{RQ}{\sin(60)} = 6$ so the radius of (PQR) is 3. The distance from O to RQ is

$\sqrt{7^2 - \left(\frac{3\sqrt{3}}{2}\right)^2} = \frac{13}{2}$ by the Pythagorean theorem and the distance from I to PQ is $\frac{3}{2}$. Therefore, $OI = 5$ and OPI is a 3 – 4 – 5 Pythagorean triple.

If we let $OI \cap RQ = M$; $QR \cap AB = C$, note that CMO and IPO are similar right triangles. Therefore, $OI = \frac{13}{2}$ means $CO = \frac{65}{8} \Rightarrow CP = CO - 4 = \frac{33}{8}$

If we let PI intersect QR at X , we see that CPX is also a 3 – 4 – 5 triangle meaning $PX = \frac{44}{8} = \frac{11}{2}$. The distance from P to QR is then $\frac{11}{2} \cdot \frac{3}{5} = \frac{33}{10}$. The answer is,

$$\frac{1}{2} \left(\frac{33}{10} \right) (3\sqrt{3}) = \frac{99\sqrt{3}}{20}$$

Problem 18 (20SIME13)

We angle chase to see that,

$$\angle IYX = \angle IAX = 90 - \angle AIX = 90 - \left(\angle \frac{A}{2} + \angle \frac{C}{2} \right) = \frac{\angle B}{2}$$

Similarly, $\angle IXY = \angle IAY = \angle \frac{C}{2}$. As a result, $XY \parallel BC$. We let $MI \cap BC = M_A$ the midpoint of BC by similar triangles.

We let $BC = x$. Note that,

$$\frac{IX}{XC} = \frac{IX}{AX} \cdot \frac{AX}{IX} = \tan\left(\frac{b}{2}\right) \tan\left(\frac{c}{2}\right)$$

Since $\tan\left(\frac{b}{2}\right) = \frac{r}{s-b}$ and $\tan\left(\frac{c}{2}\right) = \frac{r}{s-b'}$,

$$= \frac{r^2}{(s-b)(s-c)}$$

Using $A = \sqrt{s(s-a)(s-b)(s-c)} = rs$, we have $r = \sqrt{\frac{(s-a)(s-b)(s-c)}{s}}$. Therefore,

$$= \frac{s-a}{s}$$

We have,

$$\frac{IX}{XC} = \frac{s-a}{s} \Leftrightarrow \frac{CX}{CI} = \frac{s}{a} \Leftrightarrow \frac{MM_A}{M_AI} = \frac{s}{a} = \frac{41+x}{2x}$$

Let $AI \cap BC = D$. We may use angle bisector to get $DB = \frac{21}{41}x$ meaning $DM_A = \frac{1}{82}x$. By similar triangles, $\frac{PM_A}{DM_A} = \frac{41+x}{2x}$. Therefore, $PM_A = \frac{41+x}{164} = \frac{1}{4} + \frac{x}{164}$. We have,

$$\begin{aligned} 11 = BP &= M_AB - PM_A = \frac{x}{2} - \left(\frac{1}{4} + \frac{x}{164}\right) \\ \Leftrightarrow \frac{45}{4} &= \frac{81}{164}x \\ \Leftrightarrow x &= \frac{205}{9} \end{aligned}$$

39 clubs; 10 hours

Problem 20 (15AIMEII15)

Directed angles in use.

Let O_1, O_2 be the centers of P, Q respectively. Note that,

$$\angle DO_1A = 2\angle DBA = 2\angle ACE = \angle AO_2E$$

Therefore, DO_1A and AO_2E are similar triangles since $DO_1 = AO_1$ and $EO_2 = AO_2$. We have,

$$\frac{DA}{AE} = \frac{O_1A}{O_2A} = \frac{1}{4}$$

We may extend $P = O_1O_2 \cap BC$. Notice that PO_1B and PO_2C are similar so $PO_1 = \frac{1}{3}(5) = \frac{5}{3}$. As a result, PO_1B is a $3-4-5$ triangle. We may use law of cosines on BO_1A and AO_2C to get $BA = \frac{4\sqrt{5}}{5}$ and $AC = \frac{8\sqrt{5}}{5}$. We also find $BC = 4$. Therefore, BAC is a right triangle.

$$[ABD] = [ACE]$$

$$\Leftrightarrow AD \cdot AB \cdot \sin(\angle BAD) = AC \cdot AE \cdot \sin(\angle CAE)$$

Since $\angle BAC = 90$, we know $\angle BAD, \angle CAD$ are complementary. Therefore,

$$\Leftrightarrow AD \cdot AB \cdot \sin(\angle BAD) = AC \cdot AE \cdot \cos(\angle BAD)$$

Plugging in the ratios,

$$\sin(\angle BAD) = 8 \cos(\angle BAD)$$

Therefore,

$$\sin^2(\angle BAD) + \cos^2(\angle BAD) = 1$$

$$\Leftrightarrow \frac{65}{64} \sin^2(\angle BAD) = 1$$

$$\Leftrightarrow \sin(\angle BAD) = \frac{8}{\sqrt{65}}; \cos(\angle BAD) = \frac{1}{\sqrt{65}}$$

Furthermore,

$$\sin(\angle O_1AB) = \frac{\sqrt{5}}{5}; \cos(\angle O_1AB) = \frac{2\sqrt{5}}{5}$$

Therefore,

$$\begin{aligned} \sin(\angle DAO_1) &= \sin(\angle DAB - \angle O_1AB) = \frac{8}{\sqrt{65}} \cdot \frac{2\sqrt{5}}{5} - \frac{1}{\sqrt{65}} \cdot \frac{\sqrt{5}}{5} = \frac{15}{5\sqrt{13}} = \frac{3}{\sqrt{13}} \\ \Rightarrow \cos(\angle DAO_1) &= \frac{2}{\sqrt{13}} \end{aligned}$$

As a result,

$$DA = 2 \cdot 1 \cdot \cos(\angle DAO_1) = \frac{4}{\sqrt{13}}$$

Now plugging this back into the area,

$$[ABD] = \frac{1}{2} \left(\frac{8}{\sqrt{65}} \cdot DA \cdot BA \right) = \frac{1}{2} \left(\frac{8}{\sqrt{65}} \cdot \frac{4}{\sqrt{13}} \cdot \frac{4\sqrt{5}}{5} \right) = \frac{64}{65}$$

Problem 21 (14AIME15)

$$\angle DBE = 90 \Leftrightarrow \angle DGE = \angle DFE = 90$$

We let $DG = 3x \Leftrightarrow EG = 4x \Rightarrow DE = 5x \Leftrightarrow DF = EF = \frac{5x}{\sqrt{2}}$. Applying Ptolemy, we get $GF = \frac{x}{\sqrt{2}}$. We see that,

$$\angle BAC = \angle GDE = \angle GFE = \angle AFE$$

We have $AE = EF$ from AEF being isosceles. Furthermore, since $\cos(\angle GDE) = \frac{3}{5}$, we have $\cos(\angle AFE) = \frac{3}{5}$ also meaning $AF = \frac{6x}{\sqrt{2}}$. We now use Power of the Point A ,

$$\frac{6x}{\sqrt{2}} \cdot \frac{7x}{\sqrt{2}} = \frac{5x}{\sqrt{2}} \cdot 3$$

$$\Leftrightarrow x = \frac{15 \cdot \sqrt{2}}{42} = \frac{5\sqrt{2}}{14}$$

For our answer, we have $DE = 5x = \frac{25\sqrt{2}}{14}$.

45 clubs; 12 hours

Problem 23 (12AIME13)

Let O be the centers. We know $AO = 3$; $CO = 5$; $BO = 4$. For the area to be maximized, we may check that we must have O in the interior of ABC (not treating this as an actual Olympiad question so skipping details intentionally). Consider a rotation of O about A of 60 degrees to a point O' on the opposite side of AC as B . Note that $AO = AO' = 3$ and $\angle OAO' = 60$ meaning $OO' = 3$. Furthermore, $BO = CO' = 4$ while $CO = 5$ means $\angle CO'O = 90$. As a result, $\angle AO'C = 150$.

Using law of sines, $AC^2 = 25 + 12\sqrt{3}$. We see that $[ABC] = \frac{\sqrt{3}}{4}AC^2 = \frac{25}{4}\sqrt{3} + 9$.

Problem 24 (24OIME14)

Let A, B, C be the vertices of the equilateral triangle in its the original state in counterclockwise order. Let A_1, A_2 be the vertices of the pink triangle on BC in the order B, A_1, A_2, C . Similarly define $B_1, B_2; C_1, C_2$. Let O be the center of the equilateral.

By symmetry, $OA_1 = OB_1 = OC_1 := b$; $OA_2 = OB_2 = OC_2 := c$. Therefore, B_2OB_1, B_2OC_1 and analogous triangles are congruent to each other. Let $A_1A_2 = A_2B_1 = \dots = e$.

From the second diagram, we see that $BA_1 = b + 2$; $BC_2 = c - 3$ and analogous. The side length of ABC is $b + e + c - 1$ while the side length of the triangle in the second diagram is $b + e + c$. The difference in area of the two triangles would be the area of the hexagon, $\frac{7^2\sqrt{3}}{4} - 3 \cdot \frac{2^2\sqrt{3}}{4} = \frac{37\sqrt{3}}{4}$. Therefore, $b + e + c = 19$.

However, by law of sines on BC_2A_1 and OC_2A_1 ,

$$b^2 + c^2 - bc = e^2$$

$$(b + 2)^2 + (c - 3)^2 - (b + 2)(c - 3) = b^2 + 7b + c^2 - 8c - bc + 19 = e^2$$

Subtracting, we have $7b - 8c + 19 = 0 \Leftrightarrow 19 = 8c - 7b$. Furthermore,

$$b^2 + c^2 - bc = e^2$$

$$\Leftrightarrow b^2 + c^2 - bc = (19 - b - c)^2$$

$$\Leftrightarrow b = \frac{-38c + 361}{-3c + 38}$$

We may solve to get $c = \frac{57}{8}$; $b = \frac{38}{7}$. The sum is $\frac{703}{56}$.

57 clubs; 13 hours

Problem 26 (24OIME15)

We use the generalized parabola equation,

$$(ax + by + f)^2 = cx + dy + e$$

(We may check that $f = 0$ has no solution for a, b using similar process as below)

Since $f \neq 0$, we may WLOG scale a, b, c, d, e such that,

$$(ax + by + 1)^2 = cx + dy + e$$

Taking implicit differentiation,

$$2(ax + by + 1) \left(a + b \cdot \frac{dy}{dx} \right) = c + d \cdot \frac{dy}{dx}$$

Plugging in $x = 1; y = 0; \frac{dy}{dx} = 0$,

$$2(a + 1)(a) = c$$

Similarly, we may get,

$$2(ax + by + 1) \left(a \cdot \frac{dx}{dy} + b \right) = c \cdot \frac{dx}{dy} + d$$

From plugging in $x = 0; y = 3; \frac{dx}{dy} = 0$,

$$2(3b + 1)(b) = d$$

The equation is,

$$(ax + by + 1)^2 = 2(a + 1)(a)x + 2(3b + 1)(b)y + e$$

Plugging in $(1, 0)$ and $(0, 3)$ which are on the graph,

$$(a + 1)^2 = 2(a + 1)(a) + e$$

$$\Leftrightarrow (a + 1)(1 - a) = e$$

$$(3b + 1)^2 = 2(3b + 1)(b)(3) + e$$

$$\Leftrightarrow (3b + 1)(1 - 3b) = e$$

If a, b have the same sign, the axis of symmetry would have a negative slope. Therefore, a, b have opposite signs. We may check that,

$$(a + 1)(1 - a) = (3b + 1)(1 - 3b) \Leftrightarrow a = -3b$$

The equation is now,

$$(-3bx + by + 1)^2 = 2(-3b + 1)(-3b)x + 2(3b + 1)(b)y + (3b + 1)(1 - 3b)$$

If we expand, we see that this is homogenous. Therefore, plug in $b = 1$,

$$\begin{aligned} (-3x + y + 1)^2 &= 2(-2)(-3)x + 2(4)(1)y + 4(-2) \\ &\Leftrightarrow 9x^2 + y^2 - 6xy - 18x - 6y + 9 = 0 \end{aligned}$$

(I was lost on what to do here so had to read the OTIS solution)

We transform the plane by $\begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix}$. Now, we may express each point (x, y) as $X = 3x - y; Y = x + 3y$. Rewriting the previous equation,

$$\begin{aligned} &\Leftrightarrow X^2 - \frac{24}{5}X - \frac{18}{5}Y + 9 = 0 \\ &\Leftrightarrow Y = \left(X - \frac{12}{5}\right)^2 + \frac{9}{10} \end{aligned}$$

In the new coordinate system, the vertex is $\left(\frac{12}{5}, \frac{9}{10}\right)$. We reverse the change of basis, the inverse of $\begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix}$ is $\frac{1}{3 \cdot 3 - 1 \cdot (-1)} \begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 0.3 & 0.1 \\ -0.1 & 0.3 \end{pmatrix}$. The answer is given by,

$$\begin{pmatrix} 0.3 & 0.1 \\ -0.1 & 0.3 \end{pmatrix} \begin{pmatrix} 12/5 \\ 9/10 \end{pmatrix} = \begin{pmatrix} 0.81 \\ 0.03 \end{pmatrix}$$

The answer is $x = 0.81; y = 0.03$.

62 clubs; 16 hours